The documentation and process conversion measures necessary to comply with this amendment shall be completed by 7 Jul 92

INCH-POUND

MIL-S-19500/474D
AMENDMENT 3
5 April 1992
SUPERSEDING
AMENDMENT 2
6 September 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, MULTIPLE DIODE ARRAYS,
TYPES 1N5768, 1N5770, 1N5772, 1N5774, AN6100, 1N6101, 1N6496,
1N6506, 1N6507, 1N6508, 1N6509, 1N6510, AND 1N6511
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/474D, dated 6 September 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

1.5, for types 1N5768-1N6509, I_F = 100 mA dc column: Following "V_{F1}", add "1/". For column with I_F = 500 mA dc, delete "V_{F1}" and substitute "V_{F2}".

Footnote 1/: Following "300 µs", add "±50 µs".

PAGE 7

FIGURE 3, note 8, delete in its entirety and substitute the following:

"8. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied. Pointed or round lead ends are allowed."

PAGE 9

FIGURE 5, symbol D, min inches column: Delete ".272" and substitute ".255"; min millimeters column: Delete "6.92" and substitute "6.48".

FIGURE 5, following note 5, add:

"6. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5	5
6, *6a, *6b, 6c	6

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PAGE 11

3.3, Delete "and 5" and substitute "5, and 6".

* Add the following new paragraph:

"3.3.2 Die plating and mounting. Pure glass shall not be used for device mounting. Metal glass die mounting is acceptable with qualifying activity approval for die with area greater than 1000 square mils."

PAGE 12

* 4.3, on table, Screen (see table II of MIL-S-19500) requirement number 13 for JANTX and JANTXV levels, delete "Subgroups 2 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = ±25 mV dc of initial reading." and substitute "Subgroup 2 of table I herein; ΔI_{R1} = 100% of initial reading or ±25 nA dc, whichever is greater. ΔV_{F1} = ±30 mV dc of initial reading."

PAGE 13

4.5.6, delete a and b in its entirety and substitute the following:

"a. For types 1N5772, 1N5774, 1N6496, 1N6508, and 1N6509, the bridging current shall be measured by supplying the forcing function to every other interconnect pin and measuring the remaining interconnect pins (excluding common anode and common cathode pins), I_{Rbr}. Repeat the test, reversing the polarity of the forcing function.

"b. For types 1N6100, 1N6101, 1N6511, and 1N6510, the isolation current shall be measured by applying the forcing function to every other diode (anode and cathode simultaneously) and measuring the remaining diodes (anode and cathode simultaneously), I_{Rbr}. Repeat the test, reversing the polarity of the forcing function.

"c. For types 1N5774, 1N6496, and 1N6509, the isolation current shall be measured between the individual circuits by applying the forcing function to the anode and cathode of one circuit and measuring to the anode and cathode of other circuit, I_{R1}. Repeat the test, reversing the polarity of the forcing function.

"d. For types 1N5768 and 1N6506, the forcing function shall be applied to every other anode and measured on the remaining anodes, I_{Rbr}. Repeat the test, reversing the polarity of the forcing function.

"e. For types 1N5770 and 1N6509, the forcing function shall be applied to every other cathode and measured on the remaining cathodes, I_{Rbr}. Repeat the test, reversing the polarity of the forcing function."

PAGE 14

TABLE I, subgroup 2, breakdown voltage, two places: Delete "100 ms" and substitute "100 ms max"; forward voltage: Delete "PW = 300 μs" and substitute "PW = 300 μs ±50 μs".

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TABLE I, subgroup 4, forward recovery voltage (peak) tests, delete in its entirety and substitute the following:

Forward recovery voltage (peak)	4026	PW ≥ 150 ns; R_s = 50Ω; t_r = 10 ns; duty cycle ≤ 2%; (see 4.5.3)			V_{F4}			
1N5768		I_F = 500 mA					5	V(pk)
1N5770								
1N5772								
1N5774								
1N6496								
1N6506								
1N6507								
1N6508								
1N6509								
1N6100		I_F = 100 mA					5	V(pk)
1N6101								
1N6510								
1N6511								

TABLE I, subgroup 4, forward recovery time test, delete in its entirety and substitute the following:

Forward recovery time	4026	PW ≥ 150 ns; R_s = 50Ω; Duty cycle ≤ 2%, t_r = 10 ns; V_{fr} = 1.1 V_F; I_F = 500 mA dc				t_{fr}		
1N5768							40	ns
1N5770								
1N5772								
1N5774								
1N6496								
1N6506								
1N6507								
1N6508								
1N6509								
1N6100		I_F = 100 mA dc;					15	ns
1N6101								
1N6510								
1N6511								

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PAGE 16

TABLE I, subgroup 4, max limit: Delete "5" and substitute "10".

TABLE I, subgroup 4, max limit: Delete "10" and substitute "5".

TABLE I, subgroup 7, inspection column, reverse current and forward voltage: Add "except omit for the following devices".

PAGE 17

TABLE I, subgroup 7, isolation current, delete in its entirety and substitute the following:

Inspection	Method	Conditions	LTPD	Symbol	Min	Max	Units
Subgroup 7 - Cont							
Bridging current (all devices)	4016	DC method; V_R = +40 V dc and -40 V dc (see 4.5.6a through 4.5.6d)		I_{Rbr}		0.8	µA dc
Isolation current	4016	DC method; V_R = +40 V dc and -40 V dc (see 4.5.6c)		I_{Ri}		0.8	µA dc
1N5774 1N6496 1N6509							

"

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TABLE IIa, subgroup 1, conditions column: Delete "and 5" and substitute "5, and 6".

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TABLE III, subgroup 1, conditions column: Delete "and 5" and substitute "5, and 6".

TABLE III, subgroup 2, conditions column: Delete "configuration 2" and substitute "figure 2".

PAGE 21

* TABLE IV, step 5, limits column: Delete "100% of initial reading or ±20 nA dc, whichever is greater." and substitute "100% of initial reading or ±25 nA dc, whichever is greater".

* TABLE IV, step 6, limits column: Delete"±25 mV dc of initial reading" and substitute "±30 mV dc of initial reading".

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The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI
Navy - SH
Air Force - 19, 80, 85
DLA - ES
NASA - NA

User activities:
Army - CR, SM
Navy - AS, CG, MC, OS

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1333)

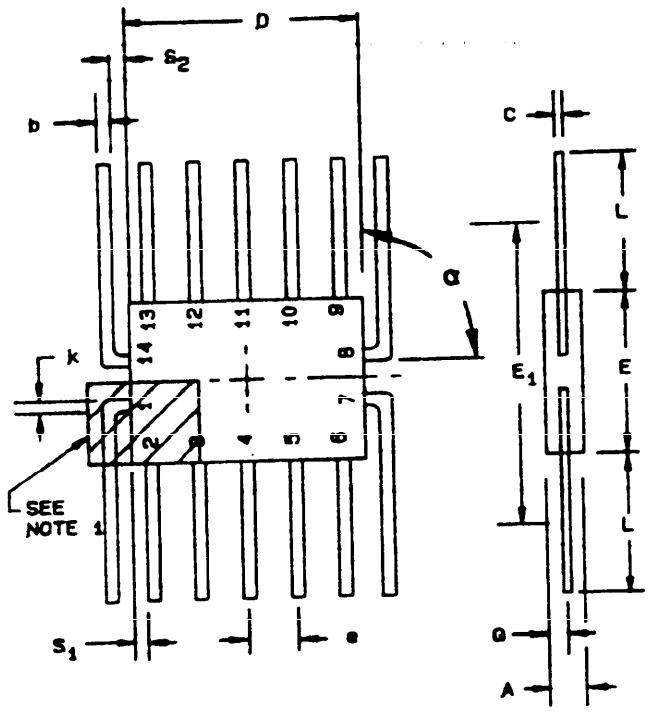
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Symbol	Inches		Millimeters		Notes	Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max			Min	Max	Min	Max	
A	.030	.090	0.76	2.29		E_3	.030		0.76		
b	.010	.019	0.25	0.48	5	e	.050 BSC		1.27 BSC		4, 6
c	.003	.006	0.08	0.15	5	k	.008	.015	0.20	0.38	10
D		.290		7.37	3	L	.240	.370	6.10	9.40	
E	.240	.260	6.10	6.60		Q	.005	.050	0.13	1.27	2
E_1		.280		7.11	3	S		.045		1.14	7
E_2	.125		3.18			S_1	.005		0.13		7, 8

NOTES:
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim. k) may be used to identify pin one. This tab may be located on either side as shown.
2. Dimensions Q shall be measured at the point of exit of the lead from the body. Dimension Q shall be .0085 (0.216 mm) minimum when lead finish A is solder.
3. These dimensions allow for off-center lid, meniscus, and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish is solder.
6. Eight spaces.
7. Applies to all four corners (lead numbers 1, 5, 6, and 10).
8. Dimension S_1 may be .000 (0.00 mm) if lead numbers 1, 5, 6, and 10 bend toward the cavity of the package within one lead's width from the point of entry of the lead into the body. For all bottom-brazed or side-brazed configurations, dimension S_1 shall be measured from the edge of the furthest extension of the metal pad or lead.
9. Optional configuration. If this configuration is used, no organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
10. Optional, see note 1 . If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.

FIGURE 1. Physical dimensions for types 1N5768, 1N5770, and 1N5772.

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Symbol	Inches		Millimeters		Notes
	MIN	MAX	MIN	MAX	
A	.030	.095	0.76	2.16	
b	.010	.019	0.25	0.48	5
c	.003	.006	0.08	0.15	5
D		.280		7.11	3
E	.240	.260	6.10	6.60	
E_1		.280		7.11	3
e	.050 BSC		1.27 BSC		4, 6
k	.008	.015	0.20	0.38	9
L	.250	.370	6.35	9.40	
Q	.010	.040	0.25	1.02	2
S_1	.005		0.130		7, 8
S_2	.004		0.10		10
α	30°	90°	30°	90°	11

FIGURE 2. Physical dimensions for types 1N5774 and 1N6100.

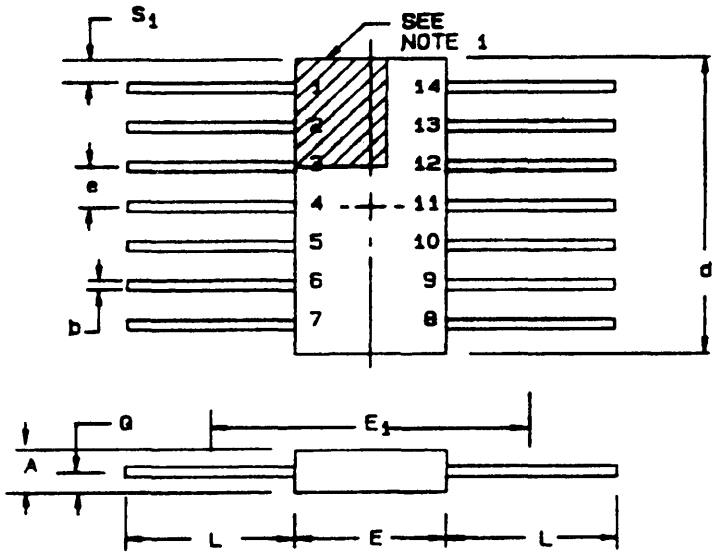
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NOTES:

1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim. k) may be used to identify pin one. This tab may be located on either side as shown.
2. Dimensions Q shall be measured at the point of exit of the lead from the body. Dimension Q shall be .0085 (0.22 mm) minimum when lead finish A is applied.
3. These dimensions allow for off-center lid, meniscus, and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied.
6. Twelve spaces.
7. Applies to all four corners (lead numbers 2, 6, 9, and 13).
8. Dimensions S_1 may be .000 (0.00 mm) if lead numbers 2, 6, 9, and 13 bend toward the cavity of the package within one lead's width from the point of entry of the lead into the body.
9. Optional, see note 1 . If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.
10. Applies to lead numbers 1, 7, 8, and 14.
11. Lead configuration is optional within dimension E except dimensions b and c apply.
12. * One year from the date of MIL-S-19500/474D, Amendment 3, this package shall become inactive and shall be replaced with the package on figure 6A.

FIGURE 2. Physical dimensions for types 1N5774 and 1N6100 - Continued.

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Symbol	Inches		Millimeters		Notes
	MIN	MAX	MIN	MAX	
A	.045	.085	1.14	2.16	
b	.010	.019	0.25	0.48	5
c	.003	.006	0.08	0.15	5
d		.390		9.91	3
E	.235	.260	5.97	6.60	
E_1		.280		7.11	3
e	.050 BSC		1.27 BSC		4, 6
L	.250	.370	6.35	9.40	
Q	.026	.045	0.66	1.14	2
S_1	.005		0.13		7, 8

FIGURE 2A. Alternate physical dimensions for types 1N5774 and 1N6100.

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NOTES
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. Dimensions Q shall be measured at the point of exit of the lead from the body. Dimension Q minimum shall be reduced by .0015 inch (0.038 mm) maximum when lead finish is solder.
3. These dimensions allow for off-center lid, meniscus and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish is solder.
6. Twelve spaces.
7. Applies to all four corners.
8. Dimensions S_1 may be .000 (0.00 mm) if lead numbers 1, 7, 8, and 14 bend toward the cavity of the package within one lead's width from the point of entry of the lead into the body.
9. Lead configuration is optional within dimension E except dimensions b and c apply.

FIGURE 2A. Alternate physical dimensions for types 1N5774 and 1N6100 - Continued.

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